A MASH 1-1 Second-Order Delta–Sigma Modulator

Theory, Quantization, Noise Shaping, Digital Cancellation, and Implementation

Educational technical note • Includes a three-level (1.5-bit) quantizer discussion

Purpose. Build an intuitive and mathematical understanding of the architecture before behavioral simulation or circuit design.

Notation. x[n] is the input; y[n] is a quantizer output; e[n] is quantization error; z■¹ denotes one sample of delay. All ideal equations are linearized models.

Abstract

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This technical note develops the intuition and mathematical model of a second-order MASH 1-1
delta–sigma modulator, with particular attention to a three-level (“1.5-bit”) quantizer. It
begins with quantization and feedback, derives the first-order noise-transfer function
(NTF), then shows how two first-order stages can be combined using digital cancellation to
obtain second-order noise shaping. It also explains oversampling, decimation, quantizer
encoding, implementation trade-offs, common failure modes, and a staged verification plan.
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The equations use an idealized linear quantization-error model. A real quantizer is nonlinear, and the exact residue path, delays, scaling, and cancellation filter depend on the chosen circuit topology. These implementation details must be derived consistently rather than copied from a generic block diagram.

1. Scope and architecture

A delta–sigma (ΔΣ) modulator converts an analog signal into a high-rate digital sequence. Instead of requiring a very fine quantizer at every sample, it combines a comparatively simple quantizer with feedback, integration, and oversampling. The output sequence carries information about the input in its average and low-frequency content. A digital low-pass/decimation filter then removes much of the high-frequency shaped quantization noise and reduces the sample rate.

MASH means multi-stage noise shaping. In a MASH 1-1 architecture, two first-order stages are cascaded. The “1-1” identifies the order of each stage; ideally, the digital cancellation network produces a second-order overall noise transfer function. This is not the same topology as a single second-order feedback loop: the MASH approach uses separate stages and cancels the first stage’s quantization-noise contribution digitally.

The main design blocks are: (1) Stage 1 integrator, quantizer, and feedback DAC; (2) a residue signal that represents Stage 1’s quantization error; (3) Stage 2 integrator, quantizer, and feedback DAC; (4) a digital cancellation filter and combiner; and (5) a final digital low-pass/decimation filter. The exact physical realization may be switched-capacitor, continuous-time, or behavioral/discrete-time.

2. Quantization: the basic error model

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A quantizer maps a continuous-valued input to one of a finite set of output levels. For
analysis, write y[n] = v[n] + e[n], where v[n] is the quantizer input, y[n] is its output,
and e[n] is the quantization error under the chosen sign convention. In an ADC context, the
input-referred error is also commonly written e[n] = x[n] − y[n]. Keep the convention
explicit: changing the sign convention changes intermediate equations, although a consistent
model gives the same physical conclusions.
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Example: if the input is 0.73 V and the represented output is 0.70 V, then e = 0.73 – 0.70 = +0.03 V. Quantization error is not eliminated by delta–sigma modulation. Feedback and integration shape where that error appears in frequency, and digital filtering suppresses the part outside the signal band.

The common linearized model treats e[n] as an additive disturbance. It is useful for deriving signal-transfer and noise-transfer functions, but it is not a complete model of a nonlinear quantizer. In particular, it does not by itself guarantee stability, absence of idle tones, or a particular SNR in a real implementation.

3. Integrator and negative feedback

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An integrator accumulates its input. A simple discrete-time accumulator is v[n] = v[n−1] +
u[n]. In a delta–sigma loop, u[n] is not usually the input alone; it is the difference
between the input and a feedback DAC signal. A representative first-order loop equation is
v[n] = v[n−1] + x[n] − d[n], where d[n] is the DAC feedback corresponding to an earlier
quantizer output. The exact sample delay depends on the circuit timing convention.
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The feedback loop continually reacts to the difference between the input and the DAC output. If the long-term average DAC output is too low, the accumulated error tends to increase and push the quantizer toward higher output levels. If the average is too high, the loop tends to push it down. Thus, individual output samples may be coarse, but the average of the sequence can represent the input accurately within the loop’s stable operating range.

The integrator provides memory. It makes persistent error matter more than a one-sample discrepancy. The quantizer produces a discrete output, and the DAC maps that output back to an analog feedback level. For a one-bit loop, the DAC often switches between two levels. For a three-level quantizer, it may use negative, zero, and positive feedback levels.

4. First-order ΔΣ model and noise shaping

For a linearized first-order modulator, a convenient canonical model is Y■(z) = X(z) + (1 – z■¹)E■(z), where X is the input, Y■ is the stage output, and E■ is its quantization error. This convention assumes unity signal transfer in the equations; a practical implementation may include a delay or gain that must be accounted for in the digital combiner.

The canonical signal transfer function (STF) is STF■(z) = 1, and the noise transfer function (NTF) is NTF■(z) = 1 – z■¹. Other timing conventions may yield STF■(z) = z■¹ while retaining the same first-order NTF. The delay is not the key noise-shaping property; the zero of the NTF at DC is.

At z = 1, corresponding to zero frequency, NTF■(1) = 0. Consequently, low-frequency quantization noise is attenuated, while noise is pushed toward higher frequencies. Near DC, |1 – e■■ω| ≈ |ω|, so the noise amplitude has a first-order slope of approximately 20 dB/decade. This describes the NTF magnitude near DC, not every detail of the output spectrum.

5. Oversampling and decimation

Oversampling means sampling substantially faster than the Nyquist rate required by the signal bandwidth. It provides a wide frequency region above the signal band into which noise can be shaped. A digital low-pass filter retains the desired low-frequency signal and attenuates out-of-band quantization noise; decimation then reduces the sample rate. The filter must suppress out-of-band energy sufficiently before downsampling to avoid aliasing it back into the signal band.

The oversampling ratio (OSR) is commonly defined as OSR = f■/(2B), where f■ is the modulator sample rate and B is the one-sided signal bandwidth, assuming a low-pass signal band from DC to B. Increasing OSR generally improves in-band noise performance, but it increases clocking and digital-processing requirements and does not fix instability, analog nonlinearity, or poor circuit noise by itself.

6. From first order to second order

A canonical second-order noise transfer function is NTF■(z) = (1 – z■¹)² = 1 – 2z■¹ + z■². Near DC, |NTF■(e■ω)| ≈ ω², giving an approximately 40 dB/decade low-frequency amplitude slope. A first-order NTF has an approximately 20 dB/decade slope. The order refers to the number of zeros at DC in the ideal NTF.

Do not infer that simply connecting two modulators in series automatically creates a second-order modulator. The stages must be arranged so the first stage’s quantization error is represented in the second stage’s input and then cancelled by a matching digital filter. The cancellation relationship, including delays and scaling, is what gives the overall second-order NTF.

7. MASH 1-1 derivation

Use a single, internally consistent canonical convention. Let the first stage be Y■(z) = X(z) + (1 – z■¹)E■(z). Let the second stage process the first stage’s residue so that Y■(z) = E■(z) + (1 – z■¹)E■(z). Here E■ is the first stage’s quantization error and E■ is the second stage’s quantization error. These are linearized analytical equations; the physical residue path must be designed to realize the assumed relationship.

The digital cancellation network forms Y(z) = Y■(z) – (1 – z■¹)Y■(z). Substituting the stage equations gives Y = X + (1 – z■¹)E■ – (1 – z■¹)[E■ + (1 – z■¹)E■]. The two E■ terms cancel, leaving Y(z) = X(z) – (1 – z■¹)²E■(z). Therefore, the ideal overall STF is unity and the overall NTF is –(1 – z■¹)²; the minus sign does not change the noise magnitude or order.

This derivation is the heart of MASH 1-1. Stage 1 supplies the signal and a shaped error term. Stage 2 processes the residue associated with E■. The digital correction removes E■’s contribution and leaves E■ shaped by a second-order difference filter. If the implementation introduces additional delays, gains, or polarity inversions, the cancellation filter must be modified to match them. A one-sample alignment error can prevent cancellation and leave a large residual noise term.

8. What “1.5-bit quantizer” means

In common ADC terminology, a 1.5-bit quantizer usually means a three-level quantizer, often with normalized output levels –1, 0, and +1. Three states require at least two binary wires/codes for representation, with one of the four two-bit codes unused in a simple encoding. The label “1.5-bit” is an engineering convention, not a literal fractional number of digital wires.

A three-level quantizer typically uses two decision thresholds and can be implemented using two comparators in a flash-style structure. Its feedback DAC must reproduce the corresponding negative, zero, and positive levels. Actual thresholds, reference levels, hysteresis, coding, and comparator polarity depend on the design specification.

A three-level quantizer is not mandatory for every MASH 1-1 architecture. Textbook MASH examples may use one-bit quantizers in both stages. A project requiring a 1.5-bit quantizer must state which stage uses it, what its thresholds and output levels are, and how the feedback DAC and digital encoding correspond to those levels. More levels can reduce quantization error or improve operating range in some designs, but they do not guarantee better performance without suitable loop scaling and stability analysis.

9. Practical architecture and design decisions

Before choosing circuit values, establish the signal bandwidth B, sampling frequency f■, OSR, input range, supply voltage, target SNR/SNDR, power budget, process or simulation environment, and whether the modulator is discrete-time or continuous-time. These determine integrator coefficients, reference levels, comparator requirements, and the decimation filter.

For a behavioral model, define the integrator state equations, quantizer thresholds, feedback levels, saturation limits, and reset conditions explicitly. For a switched-capacitor design, derive capacitor ratios and clock phases. For a continuous-time design, specify loop-filter transfer functions, DAC pulse shape, excess loop delay, and stability margins. For RTL, model the quantizer coding, sample delays, arithmetic widths, and digital cancellation; fixed-point overflow and truncation must be considered.

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A sensible development sequence is: (1) implement one first-order stage; (2) verify that its output average follows the input; (3) inspect its output spectrum and confirm first-order shaping; (4) add Stage 2 and verify its residue input; (5) implement the cancellation filter; (6) confirm the second-order NTF and in-band noise reduction; (7) sweep input amplitude and OSR; and (8) add nonidealities and verify margins.
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10. Verification plan and expected results

Use a DC input first. After startup, the average output should track the input after applying the appropriate output scaling and ignoring startup transients. Sweep several positive and negative input levels within the stable range. Check that the output does not remain saturated and that the internal integrator states stay bounded.

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Next, use a low-frequency sinusoidal input and estimate the power spectral density (PSD). A first-order stage should show low-frequency suppression consistent with an NTF zero at DC. After ideal MASH cancellation, the combined output should exhibit second-order-shaped quantization noise. Use a sufficiently long record, discard startup samples, apply a suitable window if needed, and normalize the PSD consistently before comparing results.
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Measure in-band SNR by integrating signal power and in-band noise power separately. Do not confuse SNR with SNDR: SNDR includes distortion power, while SNR excludes harmonics/distortion according to the chosen measurement convention. Compare results at the same bandwidth, sample rate, input amplitude, FFT length, and window. Verify behavior near full scale because overload and instability can invalidate the linearized model.

11. Common mistakes and debugging

Mistake 1: Treating a MASH 1-1 as a single second-order loop. Correction: it is two first-order stages plus digital cancellation.

Mistake 2: Assuming the second stage receives an abstract error signal directly. Correction: the circuit must expose or generate a physical residue that matches the mathematical error model, including gain and delay.

Mistake 3: Mixing stage equations from different delay conventions. Correction: write the exact difference equations first, transform them to z-domain, and derive the cancellation filter from those equations.

Mistake 4: Assuming a three-level quantizer automatically means higher resolution or better SNR. Correction: performance depends on the full loop, feedback DAC, scaling, and stability.

Mistake 5: Looking only at a time-domain bitstream. Correction: also inspect average output, integrator states, PSD, in-band SNR/SNDR, and behavior across input amplitudes.

Mistake 6: Believing noise shaping eliminates quantization noise. Correction: it redistributes noise in frequency; the digital low-pass filter removes much of the out-of-band component.

Mistake 7: Ignoring digital alignment. Correction: the cancellation filter must align Stage 1 and Stage 2 terms exactly. Delay mismatch, gain mismatch, residue errors, and finite precision all limit cancellation.

12. Summary

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A first-order ΔΣ loop uses an integrator and negative feedback so that its quantizer error
is shaped by NTF■(z) = 1 − z■¹. A MASH 1-1 places two first-order stages in cascade and uses
a digital cancellation filter to cancel the first stage’s quantization error. With
consistent canonical equations, the combined output is Y(z) = X(z) − (1 − z■¹)²E■(z),
yielding second-order noise shaping. Oversampling and digital decimation then suppress much
of the shaped out-of-band noise.
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For implementation, the critical work is not memorizing the block diagram. It is defining the physical residue path, quantizer levels, loop scaling, delays, feedback DAC, and cancellation arithmetic consistently, then validating the average output and spectrum in simulation before moving to circuit-level optimization.

References and further reading

1. S. R. Norsworthy, R. Schreier, and G. C. Temes, eds., Delta-Sigma Data Converters: Theory, Design, and Simulation, IEEE Press, 1997.

2. R. Schreier and G. C. Temes, Understanding Delta-Sigma Data Converters, Wiley-IEEE Press, 2005.

3. J. C. Candy and G. C. Temes, eds., Oversampling Delta-Sigma Data Converters: Theory, Design, and Simulation, IEEE Press, 1992.

These are standard references for rigorous converter theory, noise-transfer-function analysis, circuit implementation, and practical nonidealities. This note is an educational synthesis, not a reproduction of the cited books and not a claim of original research.